

PERSISTENCE TO DISEASE-MODIFYING THERAPY IN PATIENTS WITH RELAPSING MULTIPLE SCLEROSIS

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BACKGROUND

- Patients with relapsing multiple sclerosis (RMS) are maintained on disease-modifying therapy (DMT) to help slow disease progression, reduce the number of disease relapses, and limit new disease activity.¹
- Reported persistence rates to DMTs are concerningly low.^{2,3}

OBJECTIVES

- Evaluate DMT persistence over 3 years
- Assess rate and reasons for DMT discontinuation

METHODS

Design & Setting

- Retrospective review; May 2017 – Oct 2020
- Academic tertiary health system with integrated specialty pharmacy

Sample

- ≥ 3 fills of DMT (index) in May – October 2017
- Relapsing form of MS (ICD10 of G35)
- DMT prescribed by MS Clinic provider

Outcomes

- Persistence to index DMT over 3 years
- Reason for first DMT discontinuation
- Pharmacist role in first DMT discontinuation

RESULTS

TABLE 1. BASELINE CHARACTERISTICS (N=540)

Characteristic	n (%)
Age, years (median, IQR)	49 (41-57)
Gender, female	402 (74)
Race, white	453 (84)
Insurance type, commercial	331 (61)
Index DMT	
interferon	156 (29)
glatiramer acetate (branded version)	115 (21)
dimethyl fumarate (branded version)	107 (20)
fingolimod	96 (18)
teriflunomide	64 (12)

RESULTS

FIGURE 1. PROBABILITY OF DMT PERSISTENCE

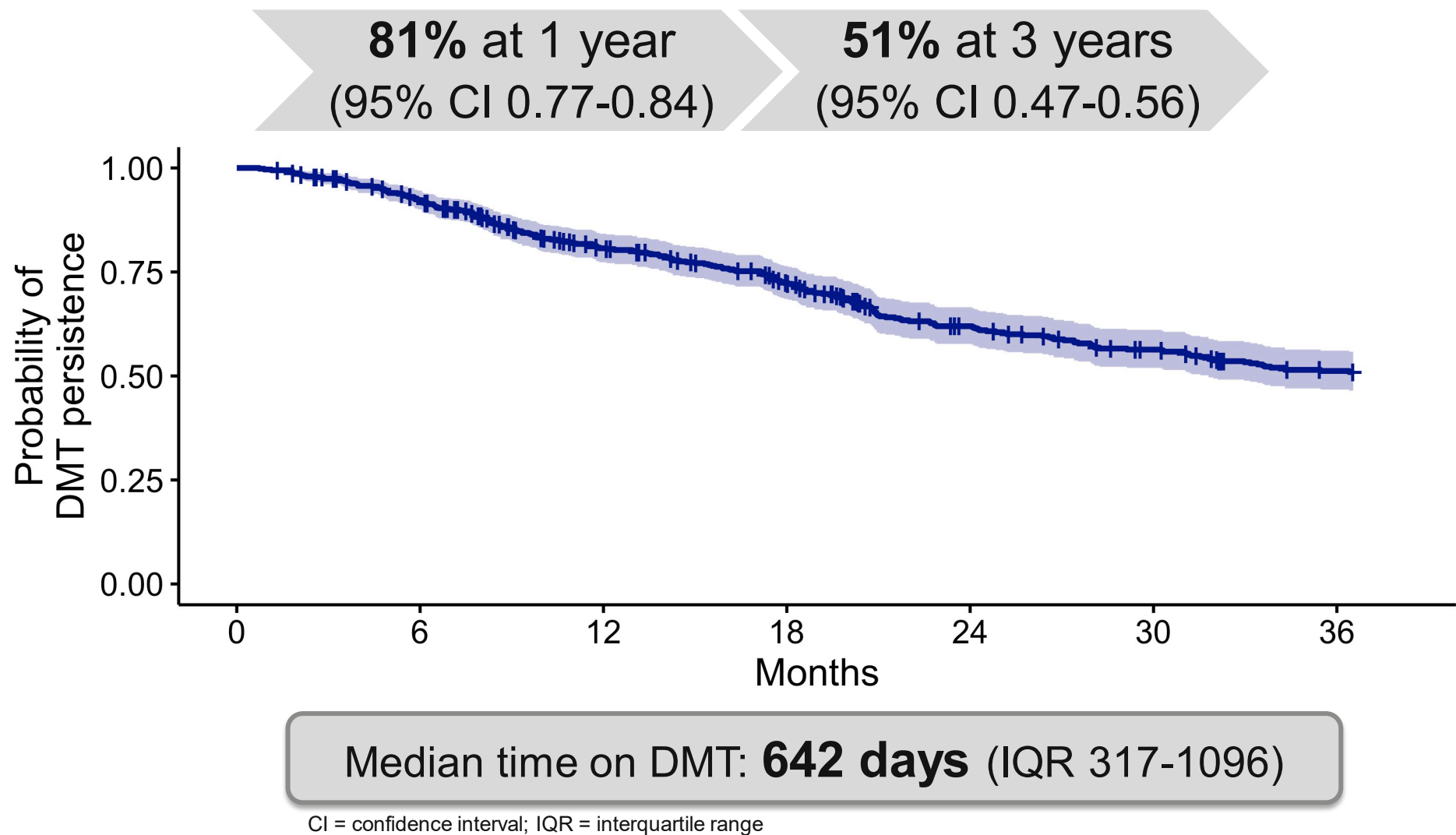


FIGURE 2. TIME TO FIRST EVENT ON DMT

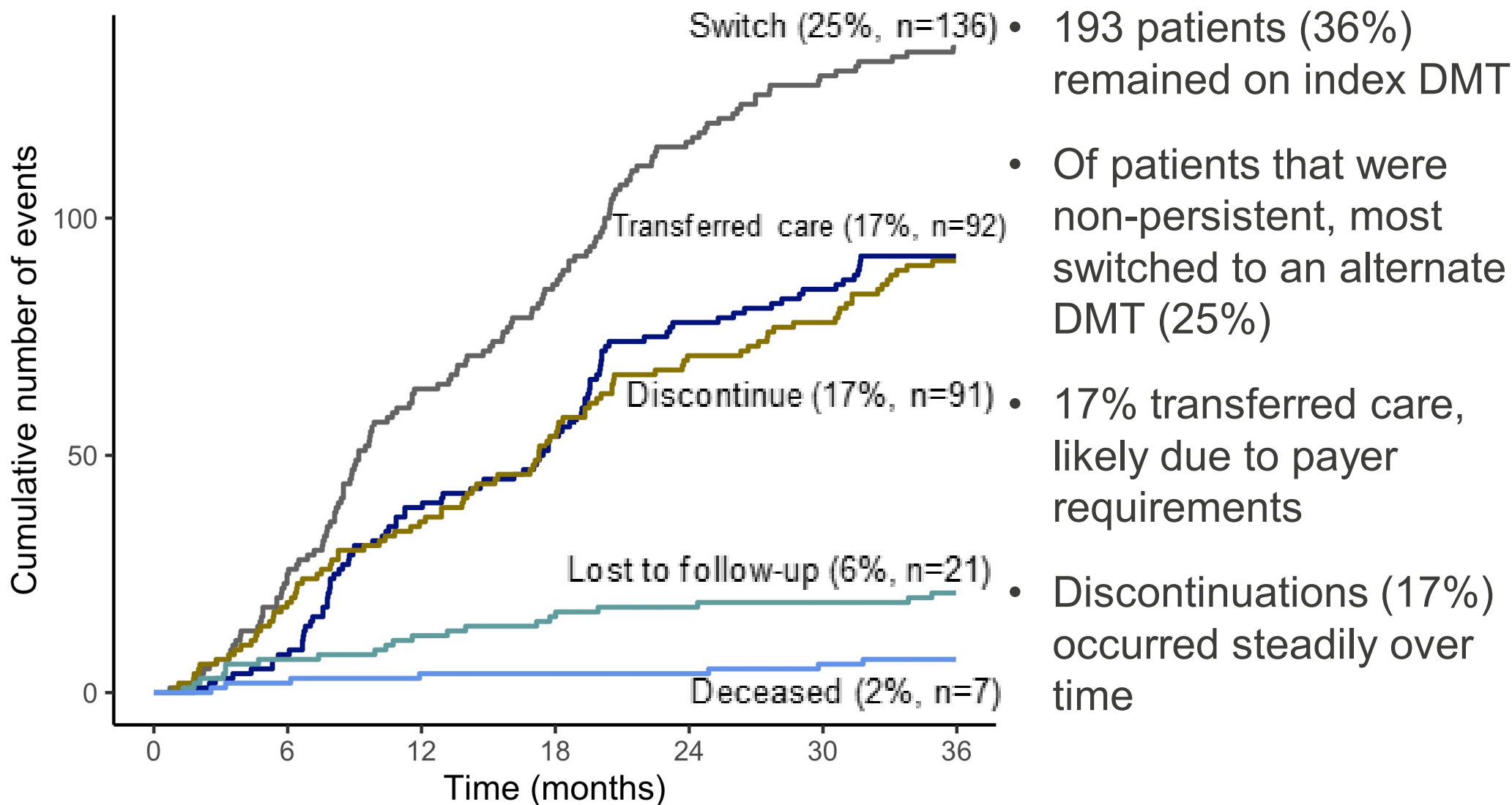
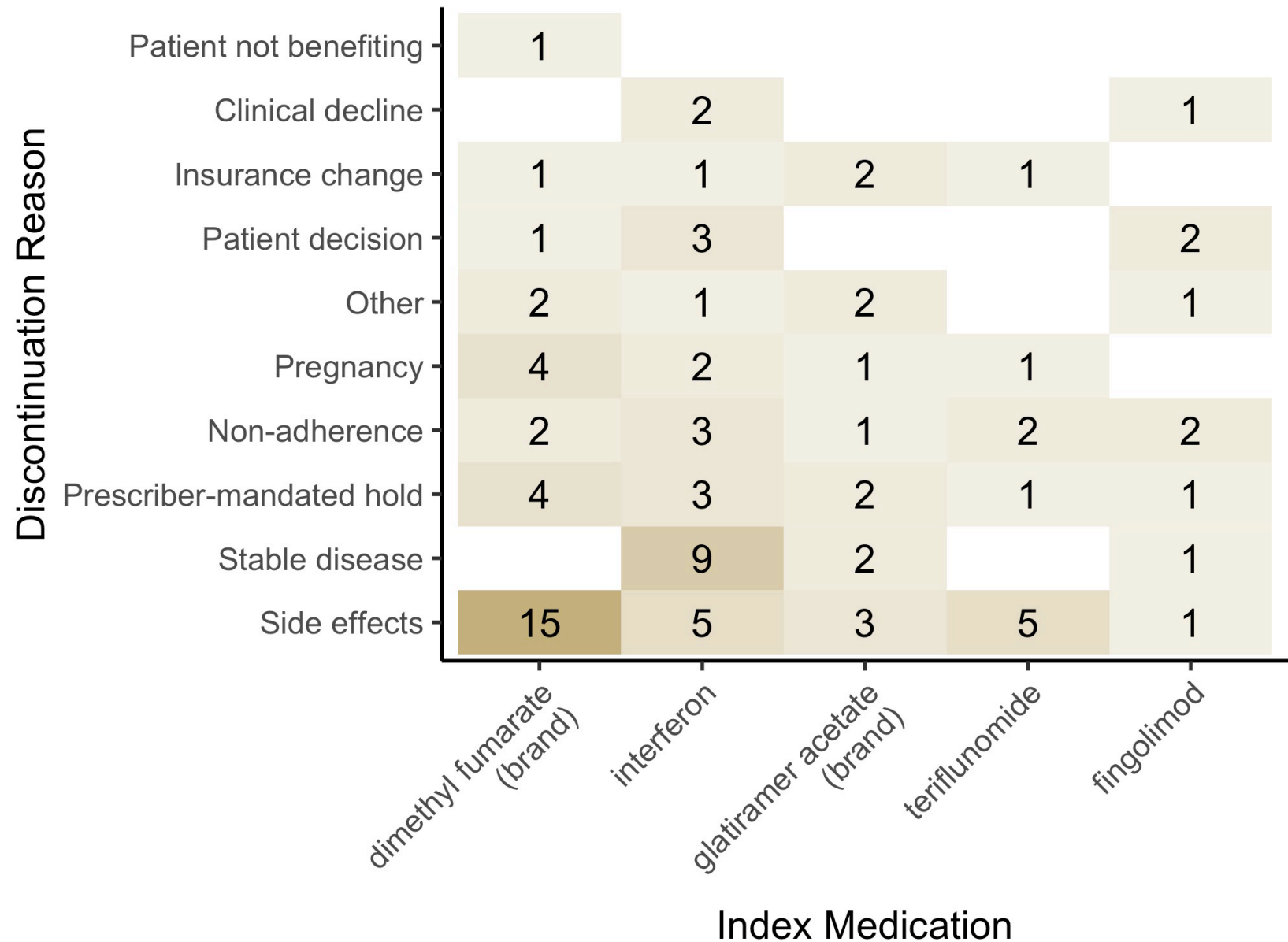


FIGURE 3. DMT DISCONTINUATION REASONS



- Dimethyl fumarate accounted for the most discontinuations (33%)
- **Side effects** were the most common discontinuation reason (32%), particularly for dimethyl fumarate
- 13% of patients discontinued due to **stable disease**, often while on interferon

SPECIALTY PHARMACIST INVOLVEMENT IN DMT DISCONTINUATIONS

- Identify the need for and initiate discontinuation due to lymphopenia, liver enzyme elevation, & pregnancy
- Communicate with the patient and prescriber
- Coordinate follow-up care, including additional lab monitoring
- Recommend new DMT option

CONCLUSIONS

- Long-term persistence to one specific DMT over 3 years is low
- Patients with RMS commonly discontinue their DMT due to side effects or stable disease
- DMT treatment changes and patient transitions occur frequently and integrated specialty pharmacists play an important role in this process